Dr. Kishan Dwivedi



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Research Interests

- Computational mechanics: Finite Element Method (FEM); Extended and Higher-Order Extended Finite Element Methods (XFEM/HOXFEM)
- Composite laminates and sandwich structures: fracture mechanics, delamination, impact and progressive damage modelling
- Composite fatigue, fatigue crack growth and cohesive-zone modelling
- Structural dynamics, vibration analysis and structural health monitoring
- Machine learning for predicting the response of damaged composite structures

Education

PhD in Mechanical Design, Indian Institute of Technology Mandi, India

- Start: 16/09/2020 | Viva/defence: 05/01/2026 | Degree awarded: 20/06/2026 | CGPA: 8.46/10
- Thesis: Fracture and Vibration Analysis of Orthotropic Composites using Higher-Order Extended Finite Element Method

M.Tech in Mechanical System Design, National Institute of Technology Srinagar, India

- 29/06/2017 – 29/11/2019 | Degree awarded: 18/02/2021 | CGPA: 8.422/10
- Thesis: Study of evolution of microstructure of 5000 h aged RAFM steel at 823 K under low cycle fatigue loading

B.Tech in Mechanical Engineering, KIT Kanpur, India

- 01/07/2011 – 28/06/2015 | Degree awarded: 22/01/2016 | 71.74%

PhD Thesis Work

Developed a higher-order extended finite element method (HOXFEM) for accurate and computationally efficient fracture and vibration analysis of cracked orthotropic composite structures. Implemented advanced crack-tip enrichment functions and variable-node elements to improve stress intensity factor predictions. Extended the framework from in-plane fracture analysis to the free-vibration response of cracked composite laminates and sandwich plates. Trained machine-learning surrogate models on HOXFEM-generated data for rapid prediction of fracture and modal responses.

Research Experience

Postdoctoral Researcher, Nazarbayev University, Kazakhstan (09/02/2026 – Present)

Project: Buckling-Resistant Double- and Multi-Layered Shells with Complex Topologies for Aerospace and Land Vehicles (BRAVE)

- Investigated the oblique impact response of double-sinusoidal corrugated-core CFRP sandwich panels using Abaqus/Explicit.
- Analysed progressive damage and delamination, and compared the impact performance of corrugated-core and honeycomb-core panels.

- Modelling tubular sandwich structures with power-law corrugated cores under axial compression, torsion and bending in Abaqus.

Visiting Doctoral Researcher, University of Alberta, Canada (04/09/2024 – 31/08/2025)

Project: Numerical investigation of unique XFEM fracture parameters for consistent global behaviour of tensile, CT and SENB specimens of pipeline steel

- Analysed multiple sets of XFEM fracture parameters that yield similar global CMOD–load behaviour, in terms of cohesive damage area, crack length and longitudinal strain.
- Proposed a criterion based on maximum longitudinal strain to identify unique sets of XFEM fracture parameters and to track the crack-tip location during crack evolution.

Junior Research Fellow, DRDO project, IIT Mandi, India (11/11/2019 – 08/09/2020)

Project: Design analysis of adhesively bonded composite patch repair of cracked aluminium aircraft panels

- Developed an XFEM code to predict fracture parameters (SIF) and fatigue life of cracked aluminium aircraft panels repaired with bonded composite patches.
- Performed an optimisation study of patch shape and size for in-situ scarf patch repair under thermo-mechanical loading.

M.Tech Project, IGCAR Kalpakkam, India (10/12/2018 – 20/06/2019)

- Performed low cycle fatigue (LCF) tests on 5000 h aged RAFM steel at 823 K, at a strain amplitude of ±0.6% and a strain rate of 3×10^{-3} s^{-1}.
- Interrupted LCF tests at 5, 10, 30 and 50% of fatigue life and studied substructure evolution using EBSD and X-ray diffraction (XRD).

Career Break (29/06/2015 – 28/06/2017)

Prepared for the Graduate Aptitude Test in Engineering (GATE); qualified GATE 2017 and was admitted to the M.Tech programme at NIT Srinagar.

Journal Publications (7)

1. **Kishan Dwivedi**, Saher Attia, Himanshu Pathak, Samer Adeeb, "XFEM fracture parameters are not unique for consistent global behavior in tensile, CT, and SENB specimen," *Engineering Fracture Mechanics*, 325, 111351, 2025. doi.org/10.1016/j.engfracmech.2025.111351
2. Siddharth Suman, **Kishan Dwivedi**, Ahmed Raza, Himanshu Pathak, "Accelerating composite sandwich plate analysis: A hybrid higher-order XFEM and ANN approach for natural frequency prediction," *Mechanics Based Design of Structures and Machines*, 53, 4711–4723, 2025. doi.org/10.1080/15397734.2025.2454351
3. **Kishan Dwivedi**, Ahmed Raza, Himanshu Pathak, Mohammad Talha, Rajat Upadhaya, "Free flexural vibration of cracked composite laminated plate using higher-order XFEM," *Engineering Fracture Mechanics*, 289, 109420, 2023. ScienceDirect
4. **Kishan Dwivedi**, Himanshu Pathak, Sachin Kumar, "Variable node higher-order XFEM for fracture modelling in orthotropic material," *Proceedings of the IMechE, Part C: Journal of Mechanical Engineering Science*, 1–25, 2023. doi.org/10.1177/09544062221148879
5. Ahmed Raza, **Kishan Dwivedi**, Himanshu Pathak, Mohammad Talha, "Free vibration of porous functionally graded plate with crack using stochastic XFEM approach," *Journal of Vibration Engineering & Technologies*, 2023. doi.org/10.1007/s42417-023-01223-w
6. Siddharth Suman, **Kishan Dwivedi**, Samanvay Anand, Himanshu Pathak, "XFEM-ANN approach to predict the fatigue performance of a composite patch repaired aluminium panel," *Composites Part C: Open Access*, 9, 100326, 2022. ScienceDirect

7. **Kishan Dwivedi**, Gaurav Arora, Himanshu Pathak, "Fatigue crack growth in CNT reinforced polymer composite," *Journal of Micromechanics and Molecular Physics*, 7, 173–184, 2022. doi.org/10.1142/S242491302241003X

Manuscripts Under Review / In Preparation (2)

1. **Kishan Dwivedi** et al., "Oblique impact performance of corrugated-double-sinusoidal-core composite sandwich panels," under review in *International Journal of Damage Mechanics* (Q1), submitted September 2026.
2. **Kishan Dwivedi** et al., "Mechanical response of tubular sandwich structures with power-law corrugated cores under axial compression, torsion and bending," in preparation (expected submission December 2026).

Conference Proceedings (4)

1. **Kishan Dwivedi**, Ahmed Raza, Himanshu Pathak, "Higher order XFEM approach for free vibration analysis of cracked composite plates," *ASME International Mechanical Engineering Congress and Exposition – India (IMECE-India 2025)*, Hyderabad, India, 10–13 September 2025, Vol. 5, V005T10A046. ASME Digital Collection
2. **Kishan Dwivedi**, Ahmed Raza, Himanshu Pathak, "Natural frequency behavior of composite laminated plates with an extended finite element approach," CDPMHM International Conference, IIT Mandi, India, June 2024; Springer Proceedings in Materials 72, pp. 169–177, 2025. doi.org/10.1007/978-981-96-5281-5_15
3. **Kishan Dwivedi**, Vani Shankar, Ajoy Kumar Pandey, K. Mariappan, Nayaka Narasaiah, Shaju K. Albert, Ghulam Ashraf Ul Harmain, "Study on cyclic softening nature of reduced activation ferritic martensitic steel through EBSD and X-ray line profile analysis," 8th International Conference on Creep, Fatigue & Creep-Fatigue Interaction (CF-8), IGCAR Kalpakkam, India, 2021. ResearchGate
4. **Kishan Dwivedi**, Himanshu Pathak, "Patch shape and size effect on performance of in-situ scarf patch repair in aircraft panel," ICRADM 2020, SVNIT Surat; *IOP Conference Series: Materials Science and Engineering*, 1004, 012002, 2020. IOPscience

Book Chapters (4)

1. **Kishan Dwivedi**, Himanshu Pathak, "Fatigue crack growth in fiber-reinforced polymer composite laminate using higher-order XFEM," in *Recent Advances in Mechanics of Functional Materials and Structures* (ACMFMS 2022), Springer LNME, 2024. doi.org/10.1007/978-981-99-5919-8_1
2. **Kishan Dwivedi**, Ahmed Raza, Himanshu Pathak, "An extended finite element framework for free vibration analysis of laminated composite plates," in *Advances in Applied Mechanics* (INCAM 2022), Springer, 2024. doi.org/10.1007/978-981-97-0472-9_5
3. **Kishan Dwivedi**, Himanshu Pathak, "Higher order enrichment functions for fracture simulation in isotropic and orthotropic material medium," in *Enriched Numerical Techniques*, Elsevier, 2024. ScienceDirect
4. Abhishek Kumar, **Kishan Dwivedi**, Himanshu Pathak, Sunny Zafar, "Material modeling and simulation approach for polymer composite," in *Lightweight Composites: Mechanics, Processing, Properties, and Applications*, Elsevier, 2024. doi.org/10.1016/B978-0-443-18852-7.00002-6

Awards and Fellowships

- SERB–University of Alberta Overseas Visiting Doctoral Fellowship 2024, for one year of collaborative research in Canada (04/09/2024 – 31/08/2025)
- MHRD Fellowship, Government of India, for the PhD programme at IIT Mandi (16/09/2020 – 16/09/2025)

- GATE Fellowship, MHRD, Government of India, for the M.Tech programme at NIT Srinagar (29/06/2017 – 28/06/2019)

Technical Skills

- **Finite element and numerical modelling:** Abaqus/CAE, Abaqus/Explicit, ANSYS APDL, DIGIMAT
- **User subroutine development:** Abaqus VUMAT for strain-rate-dependent progressive damage modelling of composites
- **Computational methods:** FEM, XFEM, higher-order XFEM, cohesive-zone modelling, progressive damage modelling, ANN-based surrogate modelling
- **Structural analysis:** impact, fracture, fatigue, free vibration and modal analysis of composite structures
- **Programming and data analysis:** MATLAB
- **Experimental and data-processing tools:** VIC-3D, Excel, OriginPro
- **CAD:** SolidWorks

Referees

Available on request.